

12.159

EE25BTECH11042 - Nipun Dasari

Question:

The determinant

$$\begin{vmatrix} 1+b & b & 1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix} \quad (0.1)$$

evaluates to

- a) 0
- b) $2b(b-1)$
- c) $2(1-b)(1+2b)$
- d) $3b(1+b)$

Solution:

Let the determinant of the given matrix be Δ .

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix} \quad (4.1)$$

We will use row operations to simplify the determinant. Applying the operations $R_1 \rightarrow R_1 - R_3$ and $R_2 \rightarrow R_2 - R_3$ does not change the value of the determinant.

First, applying $R_1 \rightarrow R_1 - R_3$:

$$\Delta = \begin{vmatrix} 1+b-1 & b-2b & 1-1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix} = \begin{vmatrix} b & -b & 0 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix} \quad (4.2)$$

Next, applying $R_2 \rightarrow R_2 - R_3$:

$$\Delta = \begin{vmatrix} b & -b & 0 \\ b-1 & 1+b-2b & 1-1 \\ 1 & 2b & 1 \end{vmatrix} = \begin{vmatrix} b & -b & 0 \\ b-1 & 1-b & 0 \\ 1 & 2b & 1 \end{vmatrix} \quad (4.3)$$

Now, we apply the column operation $C_1 \rightarrow C_1 + C_2$. This operation does not change the value of the determinant.

$$\Delta = \begin{vmatrix} b+(-b) & -b & 0 \\ (b-1)+(1-b) & 1-b & 0 \\ 1+2b & 2b & 1 \end{vmatrix} = \begin{vmatrix} 0 & -b & 0 \\ 0 & 1-b & 0 \\ 1+2b & 2b & 1 \end{vmatrix} \quad (4.4)$$

Since the first column of the matrix has two zero entries, we can easily evaluate the

determinant by expanding along the first column:

$$\Delta = 0 \begin{vmatrix} 1-b & 0 \\ 2b & 1 \end{vmatrix} - 0 \begin{vmatrix} -b & 0 \\ 2b & 1 \end{vmatrix} + (1+2b) \begin{vmatrix} -b & 0 \\ 1-b & 0 \end{vmatrix} \quad (4.5)$$

$$= (1+2b)((-b)(0) - (0)(1-b)) \quad (4.6)$$

$$= (1+2b)(0) = 0 \quad (4.7)$$

The determinant is 0, which corresponds to option (a).